No announcements this week

6: Volume in $x^2+y^2+z^2=4$, above xy-plane, below cone $z=\sqrt{x^2+y^2}$

Step 1: translate into explicit bounds

"within" $\Rightarrow$ $x^2+y^2+z^2 \le 4$

xy-plane is the plane $z=0$, so above $\Rightarrow$ $z \ge 0$

below $z=\sqrt{x^2+y^2} \Rightarrow$ $z \le \sqrt{x^2+y^2}$

Step 2: Convert to spherical

$x=\rho\cos\theta\sin\varphi$, $y=\rho\sin\theta\sin\varphi$, $z=\rho\cos\varphi$,

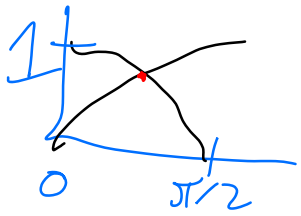
$dV = \rho^2\sin\varphi\, d\rho\, d\theta\, d\varphi$, $x^2+y^2+z^2=\rho^2$

$\rho^2 \le 4 \Rightarrow 0 \le \rho \le 2$

$0 \le z = \rho\cos\varphi \Rightarrow \cos\varphi \ge 0 \Rightarrow \varphi \in [0, \frac{\pi}{2}]$

$\rho\cos\varphi = z \le \sqrt{x^2+y^2} = \rho\sin\varphi \Rightarrow$

$\cos\varphi \le \sin\varphi \Rightarrow \varphi \in [0, \frac{\pi}{4}]$,



no constraints on $\theta \Rightarrow \theta \in [0, 2\pi]$

Step 3: setup & solve

$$V = \iiint_R dV = \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^2 \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$

$$= \int_0^{2\pi} d\theta \left(\int_0^{\pi/4} \sin\varphi \left(\int_0^2 \rho^2 d\rho \right) d\varphi \right) =$$

$$2\pi \cdot -\cos\varphi \Big|_0^{\pi/4} \cdot \frac{\rho^3}{3} \Big|_0^2 =$$

$$2\pi \left(1 - \frac{1}{\sqrt{2}}\right) \cdot \frac{8}{3} = \frac{16\pi}{3} \cdot \frac{2-\sqrt{2}}{2} = \boxed{\frac{8\pi(2-\sqrt{2})}{3}}$$

8: Evaluate by switching to spherical:

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} (x^2+y^2+z^2)^{3/2} \, dz \, dy \, dx$$

$x = \rho\cos\theta\sin\varphi$, $y = \rho\sin\theta\sin\varphi$, $z = \rho\cos\varphi$,

$dV = \rho^2 \sin\varphi \, d\rho \, d\theta \, d\varphi$, $x^2+y^2+z^2 = \rho^2$,

$2-\sqrt{\cdots} \le z \le 2+\sqrt{\cdots} \Rightarrow$

$-\sqrt{\cdots} \le z-2 \le \sqrt{\cdots} \Rightarrow$

$(z-2)^2 \le \cdots,$

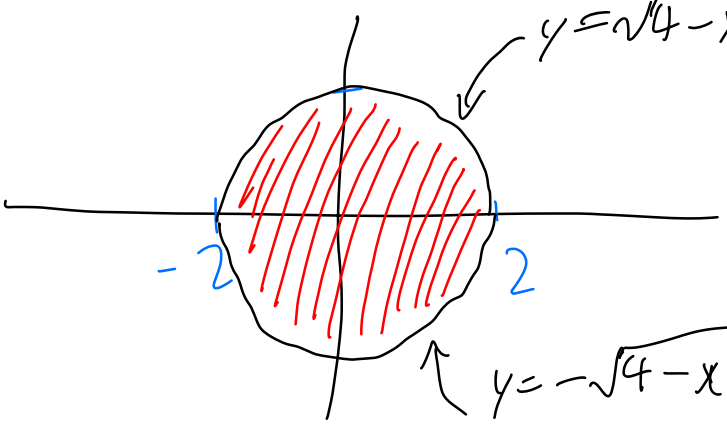
$(\rho\cos\varphi - 2)^2 \le 4 - x^2 - y^2 = 4 - \rho^2\sin^2\varphi$

$\Rightarrow \cancel{4} - 2\rho\cos\varphi + \rho^2\cos^2\varphi \le \cancel{4} - \rho^2\sin^2\varphi$

$\Rightarrow \rho^2(\cos^2\varphi + \sin^2\varphi) \le 2\rho\cos\varphi$

$\Rightarrow \rho \le 2\cos\varphi$

$-2 \le x \le 2 \quad -\sqrt{4-x^2} \le y \le \sqrt{4-x^2}$



By graphing, x & y form a perfect circle of radius 2, so the only other constraint is $4 \ge x^2+y^2 = \rho^2\sin^2\varphi$

$\Rightarrow \rho\sin\varphi \le 2.$

No θ restrictions, so $0 \le \theta \le 2\pi$

$\rho\sin\varphi \le 2$ & $\rho \le 2\cos\varphi \Rightarrow \rho \le \frac{2}{\sin\varphi}$ & $2\cos\varphi$

$\rho \le \min\left(\frac{2}{\sin\varphi}, 2\cos\varphi\right)$, but

$\frac{2}{\sin\varphi} \ge 2 \ge 2\cos\varphi$, so $\min = 2\cos\varphi$

always.

So $0 \le \rho \le 2\cos\varphi$, and

$\cos\varphi \ge 0 \Rightarrow \varphi \in \left[0, \frac{\pi}{2}\right]$

$(x^2+y^2+z^2)^{3/2}\,dz\,dy\,dx = (\rho^2)^{3/2}\rho^2\sin\varphi\,d\rho\,d\theta\,d\varphi$

$= \rho^5\sin\varphi\,d\rho\,d\varphi\,d\theta$, so at last we have

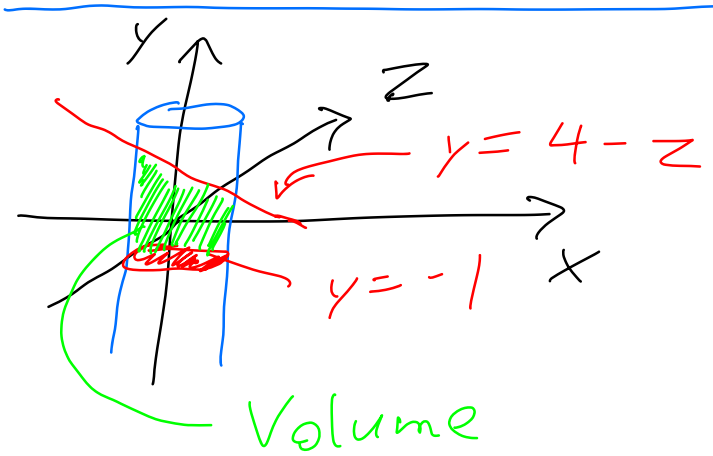
$$\int_0^{2\pi}\int_0^{\pi/2}\int_0^{2\cos\varphi} \rho^5\sin\varphi\,d\rho\,d\varphi\,d\theta =$$

$$2\pi\int_0^{\pi/2}\sin\varphi\,\frac{\rho^6}{6}\Big|_{\rho=0}^{\rho=2\cos\varphi}d\varphi =$$

$$\frac{\pi}{3}\int_0^{\pi/2}\sin\varphi\cdot\cos^6\varphi\cdot 2^6 = \frac{64}{3}\int_0^{\pi/2}-\left(\frac{\cos^7\varphi}{7}\right)'d\varphi$$

$$= -\frac{64}{21}\cos^7\varphi\Big|_0^{\pi/2} = \frac{64}{21}(0--1) = \boxed{\frac{64}{21}}.$$

2c: Volume enclosed by $x^2+z^2=4$, $y=-1$, $y+z=4$



Cylinder is in terms of x & z, so $x = r\cos\theta$, $z = r\sin\theta$, $y = y$, and $dV = r\,dr\,d\theta\,dy$.

$4 \geq x^2+z^2 = r^2 \Rightarrow 2 \geq r \geq 0$.

From graph $-1 \leq y \leq 4-z = 4-r\sin\theta$

No restrictions on θ, so $0 \leq \theta \leq 2\pi$

Now we plug everything in:

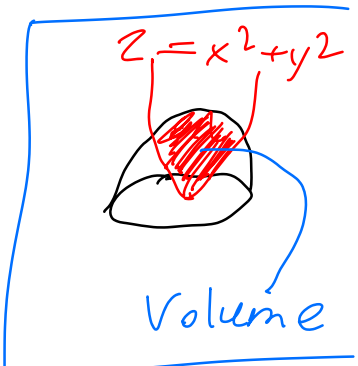
$$V = \int_E dV = \int_0^{2\pi}\int_0^{2}\int_{-1}^{4-r\sin\theta} r\,dy\,dr\,d\theta =$$

$$\int_0^{2\pi}\int_0^{2} r(5-r\sin\theta)\,dr\,d\theta = \int_0^{2\pi}\int_0^{2}(5r-r^2\sin\theta)\,dr\,d\theta =$$

$$\int_0^{2\pi}\left[\frac{5r^2}{2}-\frac{r^3}{3}\sin\theta\right]\Big|_{r=0}^{r=2} = \int_0^{2\pi}\left(10-\frac{8}{3}\sin\theta\right)d\theta$$

$= 10\theta + \frac{8}{3}\cos\theta \Big|_0^{2\pi} = \boxed{20\pi}.$

3: Setup volume inside $x^2+z^2+y^2=2$ and above $z=x^2+y^2$.



inside $\Rightarrow$ $x^2+y^2+z^2 \le 2$

above paraboloid $\Rightarrow$ $z \ge x^2+y^2$

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$, $dV = r\,dr\,d\theta\,dz$

$2 \ge x^2+y^2+z^2 = r^2+z^2$

$z \ge x^2+y^2 = r^2$

No bounds on $\theta \Rightarrow 0 \le \theta \le 2\pi$

$r^2 \le z \le \sqrt{2-r^2}$

$\Rightarrow r^2 \le \sqrt{2-r^2} \Rightarrow r^4+r^2-2 \le 0$

$\Rightarrow (r^2-1)(r^2+2) \le 0 \Rightarrow r^2-1 \le 0$

$\Rightarrow r \le 1$, so $0 \le r \le 1$

$$V = \int_0^{2\pi}\int_0^{1}\int_{r^2}^{\sqrt{2-r^2}} r\,dz\,dr\,d\theta.$$

4: Rewrite $\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{9-x^2-y^2}\sqrt{x^2+y^2}\,dz\,dy\,dx$

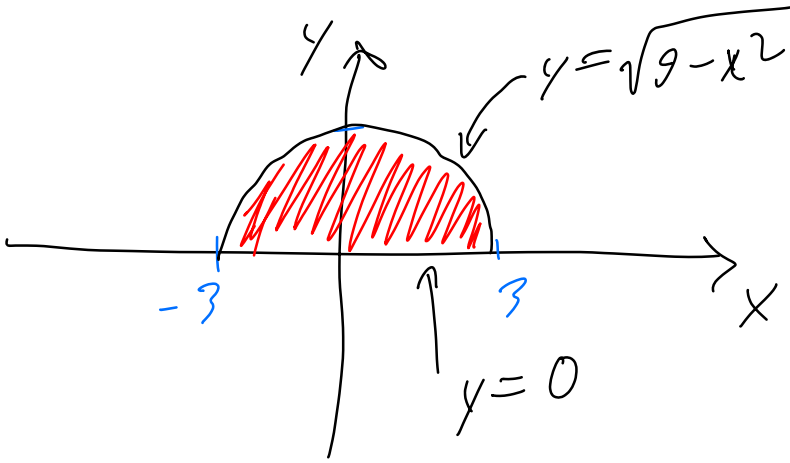
using cylindrical

x^2+y^2 exists, so ordinary cylindrical, so

$x=r\cos\theta$, $y=r\sin\theta$, $z=z$, $dV=r\,dr\,d\theta\,dz$.

$0\le z\le 9-x^2-y^2=9-r^2$

$-3\le x\le 3$ $\quad 0\le y\le\sqrt{9-x^2}$



From graph, we see that x&y bounds form top half of circle of radius 3, so $0\le r\le 3$ and $0\le\theta\le\pi$

$\sqrt{x^2+y^2}\,dz\,dy\,dx=r(r\,dr\,d\theta\,dz)=r^2\,dz\,dr\,d\theta$

So it's $\int_0^{\pi}\int_0^{3}\int_0^{9-r^2} r^2\,dz\,dr\,d\theta.$

5: Find $\iiint_E (x^2+y^2+z^2)\,dV$, $4 \le x^2+y^2+z^2 \le 9$

We see $x^2+y^2+z^2$, so let's use standard order.

$x = \rho\cos\theta\sin\varphi$, $y = \rho\sin\theta\sin\varphi$, $z = \rho\cos\varphi$,

$dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$, $x^2+y^2+z^2 = \rho^2$

$4 \le \rho^2 \le 9 \Rightarrow 2 \le \rho \le 3$ since $\rho \ge 0$.

θ, φ no restrictions, so $0 \le \theta \le 2\pi$, $0 \le \varphi \le \pi$.

$(x^2+y^2+z^2)\,dV = \rho^2(\rho^2\sin\varphi\cdots) = \rho^4\sin\varphi\,d\rho\,d\theta\,d\varphi$,

and now we substitute:

$$\int_0^{2\pi}\int_0^{\pi}\int_2^3 \rho^4\sin\varphi\,d\rho\,d\varphi\,d\theta = 2\pi\int_0^{\pi}\sin\varphi\cdot\frac{\rho^5}{5}\Big|_{\rho=2}^{\rho=3}$$

$$= \frac{2\pi}{5}\int_0^{\pi}\sin\varphi\,(243-32) = \frac{2\pi}{5}\cdot 211\int_0^{\pi}\sin\varphi\,d\varphi$$

$$= \frac{422\pi}{5}(-\cos\varphi)\Big|_0^{\pi} = \boxed{\frac{844\pi}{5}}.$$